LIGHTYEAR

MS80 Decidability calibration instrument

Status: **implemented for the documented SQL and recorded-runtime adapters**.

Delivered in PR #210.

Why it matters

Before promising to verify an unfamiliar customer estate, we need to know how much the gate can decide. The instrument measures that fraction, groups every unresolved result by cause and identifies the scope of possible normalization changes. This makes a low decision rate visible before a public engagement.

What was delivered

The calibration CLI discovers or reads a corpus, runs a supported comparison adapter and emits a portable HTML report, JSON evidence, Markdown summary and draft normalization proposals. Every unresolved unit is assigned to an explicit cause cluster. Proposals list the records they could affect so reviewers can assess their blast radius.

The tool also imports retained iDempiere comparison evidence without presenting that import as a new execution. A proposal assessment reruns the relevant comparison and measures decision changes, including previously detected divergences that would be lost. Draft proposals do not approve or automatically install equivalence rules.

Try and inspect it

```
PYTHONPATH=src python -m lightyear_calibration scan --manifest spec/calibration/sql/corpus.json --output
work/calibration/orders-baseline
PYTHONPATH=src python -m lightyear_calibration import-idempiere --report factory/idempiere-divergence-
audit/stage2-comparison.json --pairing-manifest factory/idempiere-divergence-audit/pairing-manifest.json
--output work/calibration/idempiere-baseline
PYTHONPATH=src python -m unittest tests.test_decidability -v
```

The decidability guide documents discovery, report fields, proposal assessment, report comparison and operator-selected thresholds. A configured threshold can stop pilot preparation while retaining the full report for diagnosis.

Evidence and boundaries

Decidability includes both equivalent and divergent results: finding a defect is a decision. It is not a pass rate or a promise of whole-application equivalence. The retained iDempiere Oracle-side figure is 388 decided units out of 64,414, approximately 0.6%; it must not be presented as a runtime success rate.

Tests cover accounting partitions, cause clusters, bounded proposal scope, actual assessment deltas, lost-divergence detection and malformed inputs. Unsupported adapters and unknown behavior remain explicit. Runtime comparison uses recorded observations; the instrument does not invoke an unfamiliar customer application or establish that all behaviors were exercised.

This complements services and API journey packs: the pack describes invocation, while calibration measures what the gate can decide from the resulting evidence. Customer code, representative observations and reviewed rules are still needed to qualify a particular engagement.